Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		+3		1		1		
		(ii)		$2FeAsS + 5O_2 \rightarrow As_2O_3 + Fe_2O_3 + 2SO_2$ formulae (1) balancing (1) – only awarded if **all** formulae are correct award (1) for correctly balanced equation including FeO instead of Fe_2O_3 $2FeAsS + 4\frac{1}{2}O_2 \rightarrow As_2O_3 + 2FeO + 2SO_2$		2		2	1	
	(b)	(i)		moles As_2O_3 in 100 cm^3 = $\frac{2.06}{197.8}$ = 0.0104 (1) moles As_2O_3 in 1 dm^3 = 0.0104 × 10 = 0.104 (1) concentration H_3AsO_3 = 0.104 × 2 = 0.208 mol dm^{-3} (1) accept alternative method		2	1	3	2	
		(ii)		$[H^+] = 10^{-5.11} = 7.76 \times 10^{-6}$ mol dm^{-3}		1		1	1	
	(c)			pyramidal (1) contains three bonding pairs and one lone pair of electrons (1) <u>electron pairs</u> arrange themselves around the central atom as far as possible from each other so that the <u>repulsion between them is at a minimum</u> / lp – bp repulsion > bp – bp repulsion (1) first two marks can be obtained from suitable diagram		1	2	3		